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PHARMACEUTICAL MICROBIOLOGY · PMB 201

Pharmaceutical Microbiology

A Complete Revision Guide — Bacteria, Viruses, Fungi & Parasites

PART ONE–TWO · Foundations & the Bacterial Cell

History of microbiology · cell structure & wall chemistry · the Gram stain · bacterial spores · classification

PART THREE–FIVE · Viruses, Fungi & Parasites

Viral structure & replication · HIV & coronavirus · chlamydia, rickettsia & mycoplasma · mycology & fungal drugs · introductory parasitology

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§1 SCOPE OF PHARMACEUTICAL MICROBIOLOGY

Microbiology is the science of organisms too small to be seen with the unaided eye. The word itself is built from three Greek roots — *mikros* (small), *bios* (life) and *logos* (study) — so the discipline is quite literally the study of microscopic life. Because these organisms are invisible to the naked eye, they can only be examined with the aid of a microscope. For the pharmacist the subject is far from academic: microbes both threaten the safety of medicines and supply many of the drugs we depend on, which is why it forms a core part of pharmaceutical training.

Microbiology is concerned with the form, structure, reproduction, physiology, metabolism and classification of microorganisms. It also examines how microbes are distributed in nature, how they interact with one another and with larger organisms, and the chemical and physical changes they bring about in their surroundings.

■ 1.1 Branches of microbiology

The field is usually divided according to the type of organism studied or the setting in which microbes are examined. The principal branches are summarised below.

Branch	Focus of study
Phycology	Autotrophic eukaryotic microbes, chiefly the algae.
Mycology	Fungi — yeasts, moulds and mushrooms.
Virology	Viruses, which sit on the borderline between living and non-living matter.
Protozoology	Protozoa, agents of malaria, amoebic dysentery, sleeping sickness and more.
Industrial microbiology	Use of microbes to manufacture antibiotics, vaccines, chemicals and fermented products.
Food & water microbiology	Processing, preservation and safety of food and water; control of water-borne disease.
Environmental microbiology	Role of microbes in waste breakdown and in biochemical cycles.

■ 1.2 Why microbes matter to health

Some microorganisms are decidedly beneficial — they ferment yoghurt, cheese and wine, yield penicillin, interferon and alcohol, and drive the processing of domestic and industrial waste. Others are harmful: they cause disease, spoil food, and corrode or degrade materials such as iron piping, glass lenses and timber. Microbes are also prized as models for studying the fundamentals of life, because they are tiny, grow in vast numbers in a small space, and divide extremely quickly — some bacteria double every twenty minutes, against once a day for a mammalian cell.

KEY POINT

Pharmaceutical microbiology sits at the meeting point of these two faces of microbial life: it protects medicines from contamination and spoilage while also harnessing microbes to produce the antibiotics, vaccines and other agents on which modern therapy depends.

§2 A SHORT HISTORY OF MICROBIOLOGY

Microbiology could not begin until lenses were ground fine enough to magnify organisms into view. Long before that, thinkers such as Roger Bacon (thirteenth century) had speculated that invisible creatures cause disease, but none could offer proof. The turning point came with the first reliable microscopes.

■ 2.1 Antony van Leeuwenhoek — the first observer

A Dutch cloth merchant with a passion for lens grinding, Leeuwenhoek (1632–1723) built more than 250 simple microscopes capable of magnifying 200–300 times. Peering at pond water, blood and the scrapings from his own teeth, he described tiny moving bodies he called *animalcules*. In 1683 he sketched rod-shaped, spherical and spiral forms — the first recorded observation of bacterial morphology, and a description that still holds today.

■ 2.2 Louis Pasteur — father of microbiology

Pasteur (1822–1895), a French chemist, made contributions so wide-ranging that he is often called the father of the subject. He showed that fermentation of fruits and grains into alcohol is driven by microbes, and that gently heating a liquid to about 62.8 °C for half an hour destroys the undesirable ones — the process we now call pasteurisation.

- » Using his famous swan-necked flask, he demonstrated that sterile broth stays sterile as long as airborne dust and microbes are trapped in the bend, thereby **disproving spontaneous generation**.
- » He distinguished **aerobic** from **anaerobic** life after finding that some microbes thrive only in the absence of oxygen, and named organisms that manage either way **facultative anaerobes**.
- » By ageing and weakening (attenuating) cultures, he produced protective **vaccines** against fowl cholera, anthrax and rabies — founding the principle of immunisation.

■ 2.3 Joseph Lister & Robert Koch

The English surgeon Joseph Lister (1827–1912) linked wound sepsis to airborne microbes and introduced antiseptic surgery; he also obtained the first pure culture by serial dilution. The German physician Robert Koch (1843–1910) proved that specific microbes cause specific diseases, worked out the bacillus of anthrax and of tuberculosis, and pioneered solid culture media and the streak-plate method for isolating pure colonies.

KEY POINT — KOCH'S POSTULATES

To establish that a microbe causes a disease Koch required that: (1) the organism is present in every diseased host; (2) it can be isolated and grown in pure culture; (3) the pure culture reproduces the disease in a healthy host; and (4) the same organism can be recovered from that experimentally infected host. These criteria, with modern refinements, still guide the identification of pathogens.

■ 2.4 Alexander Fleming & the antibiotic era

In 1928 Alexander Fleming noticed that a stray blue-green mould contaminating a plate of *Staphylococcus aureus* had cleared the bacteria around it. The mould, *Penicillium notatum*, was secreting a substance lethal to the bacteria; Fleming named it **penicillin**, the first of the ‘wonder drugs’ and the opening of the antibiotic age.

§3 PROKARYOTES VS EUKARYOTES

All living things are built from cells, the smallest units that can be considered alive. The deepest division in biology is not between plants and animals but between the two fundamental cell types: the **prokaryotic** cells of the bacteria, and the **eukaryotic** cells that make up animals, plants, fungi and protists. A helpful memory aid reads the roots literally — *pro-karyote* as ‘before the nucleus’ and *eu-karyote* as ‘true nucleus’.

Prokaryotes are simple, single-celled organisms without a true nucleus or membrane-bound organelles; most of their DNA lies in a central region called the **nucleoid** as a single closed loop, the circular chromosome. Eukaryotes, by contrast, package their DNA into linear chromosomes inside a membrane-bound nucleus and carry an elaborate system of internal organelles and a cytoskeleton.

Feature	Prokaryotes (bacteria)	Eukaryotes
Nucleus	Absent — DNA lies free in the nucleoid	Present, membrane-bound
Genome	Usually one circular chromosome	Several linear chromosomes
Organelles	No membrane-bound organelles	Mitochondria, ER, Golgi, etc.
Typical size	0.1–5.0 µm diameter	10–100 µm diameter
Division	Binary fission	Mitosis / meiosis
Sexual reproduction	Absent	Usually present

NOTE

The plasma membrane of a prokaryote does much of the work that dedicated organelles perform in a eukaryote — transport, biosynthesis and energy generation all take place at or within it.

§4 MORPHOLOGICAL (SURFACE) FEATURES

The outer, or morphological, features of a bacterium give it its shape and let it interact with its surroundings — moving, attaching, resisting osmotic stress and evading host defences. The most important surface structures are gathered below.

Feature	Description	Key functions
Cell wall	Rigid peptidoglycan layer outside the membrane.	Fixes cell shape; gives mechanical support and protects against osmotic lysis.
Plasma membrane	Selectively permeable phospholipid bilayer beneath the wall.	Controls transport in and out; site of respiration and ATP synthesis.
Capsule / slime layer	Gelatinous glycocalyx of polysaccharide or polypeptide.	Guards against drying and phagocytosis; aids attachment and biofilm formation.
Flagella	Long whip-like appendages driven by a basal motor.	Provide motility and chemotaxis; number and arrangement aid identification.
Pili (fimbriae)	Short hair-like protein projections, common on Gram-negatives.	Attach the cell to host tissue; sex pili mediate conjugation.

Flagella are classified by their number and position on the cell — a scheme that is itself diagnostically useful:

Arrangement	Meaning	Example
Atrichous	No flagella	<i>Corynebacterium diphtheriae</i>
Monotrichous	A single flagellum at one pole	<i>Vibrio cholerae</i>
Lophotrichous	A tuft of flagella at one pole	<i>Pseudomonas</i> spp.
Amphitrichous	Flagella at both poles	<i>Rhodospirillum rubrum</i>
Peritrichous	Flagella all over the surface	<i>Bacillus</i> spp.

§5 INTERNAL FEATURES OF THE BACTERIAL CELL

Although prokaryotes lack membrane-bound organelles, their cytoplasm contains several specialised structures that carry out the work of genetic control, protein synthesis, metabolism, storage and survival.

Structure	Description	Key functions
Cytoplasm	Semi-fluid gel of water, enzymes, salts and nutrients.	Hosts metabolic reactions such as glycolysis; suspends cell components and maintains turgor.
Nucleoid	Irregular region holding the single circular chromosome.	Carries the genetic blueprint; directs gene expression and protein synthesis.
Plasmids	Small circular DNA molecules separate from the chromosome.	Carry extra traits such as antibiotic resistance; enable horizontal gene transfer; used as cloning vectors.
Ribosomes (70S)	Granules of rRNA and protein, free in the cytoplasm.	Translate mRNA into polypeptides; make the enzymes and structural proteins the cell needs.
Inclusion bodies	Storage granules of glycogen, polyphosphate, sulfur or magnetosomes.	Hold reserves of carbon, phosphate or sulfur; magnetosomes aid orientation.
Mesosomes	Infoldings of the plasma membrane (chiefly Gram-positives).	Implicated in wall formation, DNA replication and increased enzyme surface area.
Endospores	Dormant, highly resistant bodies of <i>Bacillus</i> and <i>Clostridium</i> .	Ensure survival through heat, drought and radiation; germinate when conditions improve.

§6 CELL WALL CHEMISTRY & THE GRAM STAIN

The bacterial cell wall is built around **peptidoglycan** — a mesh of glycan chains of N-acetylmuramic acid and N-acetylglucosamine, cross-linked by short peptide bridges. This ‘chain-link fence’ encases the whole cell, fixes its shape and, crucially, lets it resist bursting in the dilute environments where bacteria live. Peptidoglycan is unique to prokaryotes, and the transpeptidase enzymes that cross-link it are a prime target of the β -lactam antibiotics such as penicillin.

■ 6.1 Other wall components

- » **Teichoic acids** — strongly negatively charged polyol-phosphate polymers linked to peptidoglycan in many Gram-positive bacteria; highly antigenic and generally absent from Gram-negatives.
- » **Lipoteichoic acids** — membrane-anchored teichoic acids that are antigenic, cytotoxic and act as adhesins (for example in *Streptococcus pyogenes*).
- » **Lipopolysaccharide (LPS)** — the major outer-membrane molecule of Gram-negatives and the source of endotoxin; it comprises a lipid A anchor, a core polysaccharide and O-antigen sugar chains.

- » **Wall-less forms** — the mycoplasmas naturally lack peptidoglycan, while L-forms are cells that have lost the ability to make it.

6.2 Gram-positive vs Gram-negative

Bacteria carry one of two wall designs. Gram-positive cells have a thick peptidoglycan layer outside the membrane; Gram-negative cells have only a thin layer, but reinforce it with a second, outer membrane that contains LPS. This difference is exactly what the Gram stain exploits.

Property	Gram-positive	Gram-negative
Peptidoglycan	Thick layer	Thin layer
Outer membrane	Absent	Present (contains LPS / endotoxin)
Stain result	Retains crystal violet — purple	Loses it, takes safranin — pink
Teichoic acids	Present	Absent

KEY POINT — THE FOUR GRAM-STAIN STEPS

(1) Flood a heat-fixed smear with the primary stain, **crystal violet**. (2) Add the mordant **Gram's iodine**, which fixes the dye in the wall. (3) **Decolourise** briefly with alcohol or acetone — the thick Gram-positive wall keeps the dye, the thin Gram-negative wall releases it. (4) **Counterstain** with safranin so the now-colourless Gram-negative cells show up pink. Because it depends on peptidoglycan, the method does not work on archaea or eukaryotes.

§7 THE BACTERIAL SPORE

A bacterial (endo)spore is a dormant, extraordinarily tough survival structure made by certain Gram-positive genera — the aerobic *Bacillus* and the anaerobic *Clostridium*. Spores let these organisms outlast conditions that would kill an ordinary vegetative cell, which is precisely why they matter in sterilisation, food safety and hospital hygiene.

7.1 Sporulation and germination

Sporulation is the programme by which a growing cell converts itself into a spore when it meets stress — starvation, extreme pH, radiation or other insults. **Germination** is the reverse: once conditions turn favourable again, the spore reactivates and returns to the vegetative state.

7.2 Layered structure

Layer (inside → out)	Composition & role
Core (protoplast)	DNA, ribosomes and enzymes held at very low water content (10–25%); rich in dipicolinic acid (DPA) and Ca^{2+} , and protected by small acid-soluble spore proteins (SASPs).
Inner membrane	Highly impermeable barrier that shields the core from chemicals.
Germ cell wall	Becomes the wall of the new vegetative cell after germination.
Cortex	Thick, loosely cross-linked peptidoglycan that keeps the core dehydrated — central to heat resistance.
Outer membrane	Protective barrier between cortex and coat.
Spore coat	Keratin-like protein layers giving chemical and enzymatic resistance.
Exosporium	Loose outermost covering (lipid, protein, carbohydrate) in some species; aids adherence.

7.3 Why spores are so resistant

- » The **low core water content** prevents heat denaturation of proteins and DNA.
- » The **DPA–Ca²⁺ complex** and DNA-binding **SASPs** stabilise the genome against heat, UV and drying.
- » **Multi-layered coats** block harmful chemicals and enzymes, and metabolic **dormancy** leaves few active targets for antimicrobials.

7.4 Resistance to specific agents

Agent	Behaviour of the spore
Moist heat	Survives boiling (100 °C) for hours; reliably killed by autoclaving at 121 °C for 15 min.
Dry heat	Needs higher temperatures — typically 160–170 °C for 1–2 hours.
Radiation	UV penetrates poorly and only partly inactivates; high-dose gamma radiation is effective.
Chemicals	Resists alcohols, phenolics and quaternary ammonium compounds; killed by aldehydes, chlorine dioxide and peracetic acid.
Desiccation / pH	Survives years of dryness and tolerates extremes of acid and alkali.

PHARMACEUTICAL & CLINICAL RELEVANCE

Spores of *Bacillus stearothermophilus* are the biological indicator used to validate autoclaves. *Clostridium difficile* spores persist on hospital surfaces and demand sporicidal cleaning; *Clostridium botulinum* spores in canned food cause botulism; and inadequate sterilisation can leave spores contaminating medicines. *Bacillus anthracis* spores are also a recognised biological weapon.

§8 CLASSIFICATION OF BACTERIA

Bacteria are grouped in many overlapping ways, each useful for a different purpose — from clinical identification to industrial selection. The most important schemes are set out here.

8.1 By shape

Cohn's classic scheme (1872) sorts bacteria into four basic shapes, with cocci further described by how the cells group together after division.

Shape	Description & arrangements	Example
Cocci	Spherical cells; occur singly, in pairs (diplococci), chains (streptococci), tetrads, cubes (sarcina) or grape-like clusters (staphylococci).	<i>Staphylococcus aureus</i>
Bacilli	Rod-shaped or cylindrical cells, single or paired.	<i>Bacillus cereus</i>
Vibrio	Short, curved, comma-shaped rods.	<i>Vibrio cholerae</i>
Spirilla	Rigid spiral forms with terminal flagella.	<i>Spirillum volutans</i>

8.2 By other physiological criteria

Basis	Groups
Nutrition	Phototrophs (energy from light) and chemotrophs (energy from chemicals); further split into litho-/organotrophs and into autotrophs and heterotrophs. Most human pathogens are heterotrophs.
Temperature	Psychrophiles (cold, optimum ≤15 °C), mesophiles (optimum ~37 °C, most pathogens), thermophiles (>45 °C) and hyperthermophiles (>80 °C).

Basis	Groups
Oxygen	Obligate aerobes, facultative anaerobes, obligate anaerobes, aerotolerant anaerobes and microaerophiles.
Flagella	Atrichous, monotrichous, lophotrichous, amphitrichous and peritrichous (see §4).
Spore formation	Spore-formers (endospore or exospore) versus non-spore-formers.
pH	Acidophiles, neutrophiles (most bacteria, pH 6.5–7.5) and alkaliphiles.
Osmotic pressure	Halophiles, extreme/obligate halophiles and facultative halophiles.

NOTE

The Gram stain (§6) is itself one of the oldest and most useful classification tools, dividing clinically important bacteria into Gram-positive and Gram-negative groups on the basis of wall structure.

§9 INTRODUCTION TO VIRUSES

Viruses are obligate intracellular parasites: a strand of nucleic acid — either DNA or RNA, never both — wrapped in a protein coat, and sometimes a lipid envelope. They have no metabolism of their own and can multiply only inside a living host cell. Inside a cell a virus shows some hallmarks of life, notably the ability to replicate; outside it is little more than an inert chemical package. Each virus has a limited host range, infecting only particular cell types.

■ 9.1 General characteristics

- » Too small to be seen with a light microscope; typically 20–300 nm across.
- » No internal organelles and no independent metabolism; they cannot grow in size.
- » Carry either DNA or RNA — which may be single- or double-stranded, linear or circular — and replicate only by commandeering host-cell machinery.

■ 9.2 Structure of the virion

An intact particle, the **virion**, has a nucleic-acid core protected by a protein **capsid**; the two together form the **nucleocapsid**. Some viruses add an outer **envelope** derived from host membrane and studded with glycoprotein spikes.

Component	Description
Nucleic acid	The genetic material — DNA or RNA, single- or double-stranded.
Capsid	Protein shell of repeating <i>capsomers</i> ; gives shape and protection.
Envelope	Host-derived lipid membrane bearing viral glycoproteins.
Spikes / peplomers	Glycoprotein projections for host attachment and antigenicity.

Capsomers are arranged symmetrically into one of two principal shapes: a rod-like **helical** capsid, or an **icosahedral** capsid — a twenty-faced solid whose low surface-to-volume ratio packs the most nucleic acid into the smallest shell. Enveloped viruses lose infectivity easily as the membrane dries, so they tend to spread through body fluids or respiratory secretions.

■ 9.3 Classification of viruses

Viruses are grouped into species, genera, families and orders — but not the higher ranks such as class, phylum or kingdom — using features such as nucleic-acid type, capsid symmetry and the presence of an envelope. A broad first split is by genetic material and disease association:

Nucleic acid	Representative viruses & diseases
DNA viruses	Poxvirus (smallpox), Herpesvirus (chickenpox, cold sores), Adenovirus (sore throat, conjunctivitis), Hepadnavirus (serum hepatitis).
RNA viruses	Orthomyxovirus (influenza), Paramyxovirus (measles, mumps), Rhabdovirus (rabies), Picornavirus (poliomyelitis), Arenavirus (Lassa fever), Retrovirus (HIV).

NOTE — SUB-VIRAL AGENTS

Two agents are even simpler than viruses. **Viroids** are tiny naked circles of RNA (300–400 bases, no protein coat) that infect plants. **Prions** are infectious proteins with no nucleic acid at all, proposed by Prusiner as the cause of scrapie and related diseases.

§10 VIRAL REPLICATION

Replication is how a virus reproduces inside a living host, which supplies most of the machinery. Every cycle follows the same broad sequence, with details set by the virus's structure and its host.

■ 10.1 Bacteriophages

Viruses that infect bacteria are called bacteriophages, or phages. The large, complex T-even phages of *Escherichia coli* have an icosahedral head of double-stranded DNA and a tail. Their growth may be **lytic** — ending in the bursting of the host — or **lysogenic**, where the phage DNA integrates quietly into the host chromosome as a **prophage**.

Stage of the lytic cycle	What happens
1. Adsorption	Tail fibres bind specific receptors on the host surface, setting host specificity.
2. Penetration	Lysozyme weakens the wall and the tail injects viral DNA; the empty capsid stays outside.
3. Replication	Host synthesis is switched off; host machinery and nucleotides are redirected to make phage DNA and protein.
4. Assembly	Heads and tails are built separately, the head is filled with DNA, then joined to the tail.
5. Release	Phage lysozyme bursts the cell, freeing 50–200 new virions (the burst size).

In the **lysogenic** cycle the injected DNA circularises and integrates into the host genome by site-specific recombination. The prophage is copied along with the host at each division until some stress — UV light, for instance — triggers **induction**, excising the phage DNA and launching the lytic cycle.

■ 10.2 Animal and retroviral replication

Animal viruses vary with their genome. A single-stranded DNA genome is first copied to double-stranded DNA before transcription. Among RNA viruses, a **positive-sense** (+ssRNA) genome acts directly as mRNA, whereas a **negative-sense** (–ssRNA) genome must first be copied to +ssRNA by a viral RNA-dependent RNA polymerase.

Retroviruses such as HIV take a different route. They carry **reverse transcriptase**, which builds a complementary DNA copy from their +ssRNA. This is made double-stranded and integrated into the host chromosome as a **provirus** — similar to a prophage, except that the provirus is never excised and can establish lifelong, chronic infection.

■ 10.3 Plant viruses

Most plant viruses carry a +ssRNA genome that behaves like mRNA. They act as biotrophic parasites, establishing infection without immediately killing the host, much as temperate phages do in bacteria.

§11 MEDICALLY IMPORTANT VIRUSES

■ 11.1 Human Immunodeficiency Virus (HIV)

HIV is a retrovirus (family *Retroviridae*, genus *Lentivirus*) that causes AIDS — a chronic, potentially fatal collapse of the immune system. It is a spherical, enveloped virus about 100–120 nm across, thought to have crossed to humans from simian immunodeficiency virus in the early 1980s. Frequent errors during reverse transcription drive rapid mutation, generating variants that resist both the immune response and antiretroviral drugs. Two strains exist: **HIV-1**, the more virulent and widespread, and **HIV-2**, less pathogenic and largely confined to West Africa.

Transmission is through specific body fluids — blood, semen, vaginal and rectal fluids and breast milk — chiefly by unprotected sex, shared needles, and from mother to child during pregnancy, delivery or breastfeeding; more rarely through unsafe transfusions or needlestick injury.

Step of the HIV life cycle	Event
Attachment	Surface gp120 binds CD4 receptors on T-helper cells, with a CCR5 or CXCR4 co-receptor.
Fusion & entry	gp41 fuses the envelope with the host membrane, admitting the viral RNA and enzymes.
Reverse transcription	Reverse transcriptase copies the RNA into complementary DNA (cDNA).
Integration	Integrase splices the viral cDNA into the host genome, forming a provirus.
Transcription & translation	Host machinery reads the provirus to make new viral RNA and proteins.
Assembly & release	New virions assemble and bud from the cell to infect fresh CD4+ cells.

HIV steadily destroys the CD4+ T lymphocytes central to immunity; as their count falls, the body becomes increasingly vulnerable to opportunistic infection — the defining feature of AIDS.

11.2 Coronaviruses

Coronaviruses are enveloped +ssRNA viruses (roughly 26–32 kb) whose crown of surface spike proteins gives the family its name. In humans they range from common colds to severe respiratory syndromes — SARS-CoV (2002), MERS-CoV (2012) and SARS-CoV-2, the cause of COVID-19 (2019). The spike (S) protein mediates receptor binding and fusion: SARS viruses bind the ACE2 receptor, while MERS-CoV binds DPP4. Spread is mainly by respiratory droplets and aerosols, with contaminated surfaces a secondary route.

KEY POINT — COVID-19 VACCINE STRATEGIES

Three main approaches all present the spike protein to the immune system: **mRNA vaccines** deliver the genetic code for spike (Pfizer, Moderna); **viral-vector vaccines** use a non-replicating adenovirus to carry the gene (AstraZeneca, J&J); and **inactivated vaccines** contain whole killed virus (Sinovac, Sinopharm).

11.3 Major DNA and RNA virus families

Family	Example(s)	Disease
Herpesviridae	HSV-1/-2, VZV, EBV, CMV	Herpes, chickenpox, shingles, mononucleosis
Papillomaviridae	Human papillomavirus	Warts, cervical cancer
Hepadnaviridae	Hepatitis B virus	Hepatitis B, liver cancer
Orthomyxoviridae	Influenza A, B, C	Influenza
Paramyxoviridae	Measles, mumps, RSV	Measles, mumps, bronchiolitis
Flaviviridae	Dengue, Zika, HCV	Dengue, Zika, yellow fever, hepatitis C
Filoviridae	Ebola, Marburg	Viral haemorrhagic fever
Rhabdoviridae	Rabies virus	Rabies

§12 CHLAMYDIA, RICKETTSIA & MYCOPLASMA

These three genera are bacteria, but each departs from the textbook model — two are obligate intracellular parasites and the third has no cell wall at all — which shapes both the disease they cause and how they are treated.

12.1 Chlamydia

Chlamydiae are obligate intracellular bacteria that lack peptidoglycan and run a distinctive two-stage life cycle. The small, tough, metabolically inactive **elementary body (EB)** is the infectious form; once inside a host cell it converts,

within a membrane-bound inclusion, into the larger **reticulate body (RB)**, which is metabolically active and multiplies by binary fission before reverting to EBs that are released to infect new cells. *Chlamydia trachomatis* is the leading infectious cause of blindness (trachoma) and the most commonly reported bacterial STI, a major cause of infertility.

■ 12.2 Rickettsia

Rickettsiae are small Gram-negative obligate intracellular bacteria spread by arthropod vectors — the bites of ticks or mites, or the faeces of lice and fleas. From the skin they spread through the blood to infect the lining of blood vessels, multiplying by binary fission in the cytosol. They cause the spotted fevers (e.g. *R. rickettsii*, Rocky Mountain spotted fever) and the typhus group (*R. prowazekii*, epidemic typhus).

■ 12.3 Mycoplasma

Mycoplasmas are unique among bacteria in having no cell wall, so they are hard to see with the Gram stain and are intrinsically resistant to wall-targeting antibiotics such as penicillins and cephalosporins. *Mycoplasma pneumoniae* causes atypical (‘walking’) pneumonia, while other species and *Ureaplasma* infect the genital tract.

Feature	Chlamydia	Rickettsia	Mycoplasma
Cell wall	No peptidoglycan	Gram-negative wall	Absent entirely
Lifestyle	Obligate intracellular	Obligate intracellular	Free of host (extracellular)
Transmission	Sexual / direct contact	Arthropod vectors	Respiratory / sexual
Wall antibiotics	Ineffective	Effective	Ineffective

§13 INTRODUCTION TO FUNGI (MYCOLOGY)

The study of fungi is called **mycology**. Fungi are eukaryotic organisms — microscopic or macroscopic — that possess a rigid cell wall, live almost everywhere, and feed as heterotrophs. Because they yield so many antibiotics yet also cause disease, they occupy an important place in pharmacy.

■ 13.1 Structure

Except for the single-celled yeasts, most fungi are filamentous: they grow as thread-like **hyphae** that weave together into a mesh called a **mycelium**. The cell wall is built of **chitin** and polysaccharides, enclosing a protoplast differentiated into membrane, cytoplasm, organelles and nuclei.

■ 13.2 Key characteristics

- » Eukaryotic, non-vascular, non-motile heterotrophs, either unicellular or multicellular.
- » Reproduce by **spores** — sexually or asexually — and show alternation of generations.
- » Lack chlorophyll, so cannot photosynthesise; they store food as starch and synthesise chitin.

■ 13.3 Are fungi closer to plants or animals?

Trait	Plants	Animals	Fungi
Cell wall	Yes	No	Yes
Sterols in membrane	No	Yes	Yes
Uses chitin	No	Some	Yes
Locomotion	Never	Almost always	Never
Nutrition	Autotroph	Heterotroph	Heterotroph

■ 13.4 Classification by nutrition

- » **Saprophytic** — feed on dead organic matter (e.g. *Rhizopus*, *Penicillium*, *Aspergillus*).
- » **Parasitic** — draw nutrients from a living host (e.g. *Taphrina*, *Puccinia*).
- » **Symbiotic** — live in mutually beneficial partnership with another species (e.g. lichens, mycorrhizae).

§14 CLASSIFICATION OF PATHOGENIC FUNGI

Group	Character	Clinically important examples
Yeasts	Unicellular, reproduce by budding, form smooth colonies.	<i>Candida</i> (esp. <i>C. albicans</i>); <i>Cryptococcus neoformans</i> (meningitis).
Moulds	Multicellular hyphae, fuzzy colonies, reproduce by spores.	<i>Aspergillus</i> ; Mucorales (<i>Mucor</i> , <i>Rhizopus</i>) — rhinosinusitis, pulmonary infection.
Mushrooms	Fleshy spore-bearing fruiting bodies.	Toxic species such as <i>Amanita phalloides</i> (death cap) — liver & kidney failure.
Dimorphic fungi	Mould in the environment, yeast in the host.	<i>Histoplasma</i> , <i>Blastomyces</i> , <i>Coccidioides</i> — subacute lung infection.
Dermatophytes	Live on keratin; cause superficial skin/hair/nail infection.	<i>Tinea</i> species — ringworm, athlete's foot.

Group	Character	Clinically important examples
Pneumocystis	Once thought a protozoan; a fungus since the 1980s.	<i>P. jirovecii</i> — Pneumocystis pneumonia (PCP).

NOTE — BACTERIA THAT MIMIC FUNGI

Nocardia and *Actinomyces* were once classed as fungi but are in fact bacteria. They cause subacute, chronic infections of lung, CNS and soft tissue — often clinically indistinguishable from fungal disease — yet respond only to antibacterials, not antifungals.

§15 FUNGAL INFECTIONS

Fungal infections (mycoses) most often strike immunocompromised patients, usually run a subacute to chronic course, and generally do not pass from person to person — with notable exceptions such as the dermatophytes. Ordinary antibacterial antibiotics do not help. They are classified by the depth of tissue involved.

Category	Site	Examples
Superficial	Outer skin, hair, nails	Tinea capitis (scalp), tinea corporis (body), tinea pedis (athlete's foot), tinea versicolor.
Mucocutaneous	Mucous membranes	Oral thrush, oesophageal candidiasis, vulvovaginal candidiasis.
Subcutaneous	Beneath the skin	Sporotrichosis ('rose-gardener's disease').
Deep (systemic)	Internal organs	Primary pulmonary histoplasmosis, blastomycosis; opportunistic cryptococcal meningitis and <i>Candida</i> endocarditis.

§16 IMPORTANCE OF FUNGI IN PHARMACY

Beyond their role in recycling, food and biocontrol, fungi are indispensable to pharmacy — above all as producers of antimicrobial drugs. The table gathers the most important pharmaceutical products derived from fungi and related organisms.

Product	Source	Use / note
Penicillin	<i>Penicillium chrysogenum</i> (<i>P. rubens</i>)	First antibiotic; kills Gram-positive bacteria (Fleming, 1928).
Streptomycin	<i>Streptomyces griseus</i>	Active against Gram-negative bacteria.
Griseofulvin	<i>Penicillium griseofulvum</i>	Antifungal for ringworm and athlete's foot.
Ergot alkaloids	<i>Claviceps purpurea</i>	Cause powerful uterine contraction; source of LSD; highly poisonous.
Ergosterol	Many moulds and yeasts	A source of vitamin D.
Riboflavin (vit. B2)	<i>Ashbya gossypii</i>	Yeast is a major source of the B-complex vitamins.
Calvacin	Giant puffball (<i>Calvatia</i>)	Anticancer substance active against stomach tumours.

KEY POINT

The single most important contribution of fungi to pharmacy is the manufacture of antimicrobials. Related actinomycetes add further agents such as chloramphenicol, aureomycin and terramycin, extending the reach of antibiotic therapy against otherwise stubborn infections.

§17 HOST-PARASITE RELATIONSHIPS

Parasitology is the branch of biology and medicine dealing with parasites and their hosts. A parasite lives on or in another organism, the host, at the host's expense. It is best understood against the wider set of symbiotic relationships.

Relationship	Definition	Example
Parasitism	One organism benefits while the host is harmed.	<i>Plasmodium</i> in humans.
Commensalism	One benefits; the other is neither helped nor harmed.	<i>Entamoeba coli</i> in the gut.
Mutualism	Both partners benefit.	Termite–protozoa cellulose digestion.

■ 17.1 Types of host

- » **Definitive host** — harbours the adult or sexually reproducing stage of the parasite.
- » **Intermediate host** — harbours the larval or asexual stage; more than one may be needed.
- » For malaria, the *Anopheles* mosquito is the definitive host (sexual stage) and the human is the intermediate host (asexual stage).

■ 17.2 Classifying parasites by habit

- » By habitat — **ectoparasites** live on the surface (lice, fleas, ticks), **endoparasites** live inside the body (*Plasmodium*, *Entamoeba histolytica*).
- » By dependence — **obligate** parasites cannot survive without a host, **facultative** ones are normally free-living but can turn parasitic when the chance arises.

§18 CLASSIFICATION OF PARASITES

The parasites of medical importance fall into two kingdoms: the single-celled **protozoa** (kingdom Protista) and the multicellular **helminths** (kingdom Animalia), alongside the **arthropods** that act as vectors.

■ 18.1 Protozoa

Protozoa are microscopic, single-celled eukaryotes divided into four classes by their means of movement: amoebae (pseudopods), flagellates (flagella), ciliates (cilia) and sporozoa (no locomotor organelle, reproducing by sporogony and schizogony).

■ 18.2 Helminths

Helminths are macroscopic multicellular worms with differentiated tissues and organ systems, ranging from under a millimetre to over a metre. They fall into three groups:

Group	Common name	Example / disease
Nematodes	Roundworms	<i>Ascaris lumbricoides</i> — ascariasis
Cestodes	Tapeworms	<i>Taenia</i> spp. — taeniasis
Trematodes	Flukes	<i>Schistosoma</i> , <i>Fasciola</i> — schistosomiasis, liver-fluke disease

■ 18.3 Arthropods as vectors

Insects and arachnids both cause disease directly and transmit other pathogens: mosquitoes, fleas and lice carry *Plasmodium*, *Rickettsia* and *Trypanosoma*, while ticks and mites transmit rickettsiae and babesia. A **biological vector** is essential to the parasite's life cycle (the malaria mosquito); a **mechanical vector** merely carries it (the housefly in amoebiasis).

§19 SOURCES & TREATMENT OF PARASITIC DISEASE

■ 19.1 Sources of infection

Source	How infection occurs
Soil	Ingesting embryonated eggs, or larvae penetrating exposed skin (hookworm).
Water	Swallowing infective forms present in contaminated water.
Food	Faecally contaminated food (amoebic cysts, helminth eggs) or undercooked meat carrying larvae (<i>Taenia saginata</i> , <i>Trichinella spiralis</i>).
Insect vectors	Biological (mosquito — malaria) or mechanical (housefly — amoebiasis) transfer.
Animal reservoirs	Cattle (beef tapeworm), dogs (hydatid disease), cats (toxoplasmosis).

■ 19.2 Antiparasitic drugs

Treatment rests chiefly on chemotherapy, occasionally supported by surgery. The main drugs of choice are summarised below.

Drug	Indication
Metronidazole / tinidazole / ornidazole	Amoebic colitis, giardiasis, trichomoniasis.
Chloroquine	Uncomplicated benign malaria.

Drug	Indication
Artemisinin derivatives, quinine, mefloquine	Complicated or falciparum malaria.
Primaquine	Relapse of vivax malaria.
Praziquantel	All cestode and trematode (tapeworm and fluke) infections.

KEY POINT

Rational antiparasitic therapy depends on correctly identifying both the parasite and its life-cycle stage: the same infection can require different drugs for the acute attack, for radical cure of dormant stages, and for prophylaxis.

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